

DAY 7

Case Studies and Final Project Presentations

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

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This content is educational material only and does not constitute financial, investment, or trading advice.

Financial modeling mastery is proven not in theory but in execution. This chapter transforms you into a practitioner who can build, stress-test, present, and defend a complete financial model under exam and real-world conditions. You will integrate DCF valuation, scenario analysis, transaction modeling, and best-practice spreadsheet architecture into a single coherent deliverable. By the final page you will know how to structure a case-study model from a blank sheet, identify and communicate key assumptions, evaluate outputs critically, and present findings with the rigor expected of a certified financial modeling and valuation analyst.

I Architecting a Case-Study Model: Inputs, Calculations, and Outputs

A well-architected financial model is not merely a calculation engine — it is a communication tool. Examiners, auditors, and senior stakeholders must be able to trace every number from its source assumption to its final output without hunting through tangled worksheets. The three-tab architecture described here enforces that discipline structurally, so transparency is built in rather than bolted on.

The Three-Tab Architecture

Every model should contain at minimum three functional sheets: Input, Calculation, and Output. A fourth Cover tab completes the package.

- Input tab — the single source of truth for all assumptions. Hard-coded values live only here. Nothing on any other sheet should contain a hard-coded number.
- Calculation tab — performs all mathematical operations by referencing the Input tab. No raw numbers; formulas and green-linked cells only.
- Output tab — presents results in a clean, reader-friendly format by pulling from the Calculation tab. No new calculations; display only.
- Cover tab — records the model name, version number, date last modified, data sources, and the name of the analyst. This satisfies both auditor requirements and exam documentation standards.

Color-Coding Convention

Adopt a consistent, industry-standard color scheme throughout:

| Color | Meaning |

|-----|-----|

| Blue font | Hard-coded input (Input tab only) |

| Black font | Formula calculated on the current sheet |

| Green font | Link pulling a value from another sheet |

Apply these colors cell by cell, not by range formatting. A reviewer scanning the sheet should instantly know whether a cell is an assumption, a formula, or a cross-sheet reference.

Labeling Assumptions

Every input cell must carry a brief comment explaining why that assumption was chosen. For example, a price assumption of \$120/user might carry the comment: "Based on current published list price; reviewed Q1 2024." This practice satisfies auditors who need to verify reasonableness and examiners who expect evidence of analytical judgment.

Avoiding Circular References

Circular references — where a cell depends on itself through a chain of references — introduce silent errors and make models unreliable. Structure your formulas so that each period's calculation depends only on prior-period outputs or Input-tab values. Enable iterative calculation settings only when a specific modeling requirement demands it (for example, interest on a revolver that feeds into EBIT), and document the reason with a clearly labeled note on the Cover tab.

Worked Example: SaaS Three-Tab Revenue Model

Scenario: A SaaS company charges \$120 per user per month. The company starts Year 1 with 1,000 users and expects 15% annual user growth. Build a five-year revenue forecast.

Key formula:

$$\text{Revenue}_t = \text{Revenue}_{\{t-1\}} \times (1 + g)$$

Step 1 — Input Tab

Cell	Label	Value	Comment
B3	Monthly price per user	\$120	Published list price, Q1 2024
B4	Year 1 users	1,000	Current contracted base
B5	Annual growth rate (g)	15%	Management guidance, conservative case

All three cells are formatted in blue font. No other sheet contains these numbers.

Step 2 — Calculation Tab

First, compute Year 1 annual revenue by linking to the Input tab:

$$\text{Year 1 Revenue} = \text{Input!B3} \times \text{Input!B4} \times 12$$

$$\text{Year 1 Revenue} = \$120 \times 1,000 \times 12 = \$1,440,000$$

Then apply the compounding formula for Years 2–5. Each cell references the prior-year cell on the same sheet (black font formula) and the growth rate via a green-font link to Input!B5:

Year	Formula	Revenue
1	=Input!B3 × Input!B4 × 12	\$1,440,000
2	=C5 × (1 + Input!\$B\$5)	\$1,656,000
3	=C6 × (1 + Input!\$B\$5)	\$1,904,400
4	=C7 × (1 + Input!\$B\$5)	\$2,190,060
5	=C8 × (1 + Input!\$B\$5)	\$2,518,569

Note the absolute reference \$B\$5 — this prevents the growth-rate link from shifting when the formula is copied down.

Step 3 — Output Tab

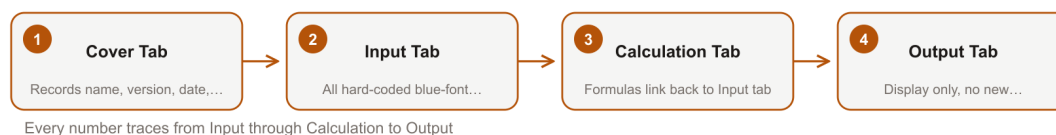
The Output tab pulls Year 1–5 revenues from the Calculation tab using green-font links and presents a summary income statement. No new arithmetic occurs here. A simple gross margin assumption (also sourced from the Input tab) can extend the display to gross profit.

Common mistake: Analysts frequently hard-code the growth rate directly into the Calculation tab — for example, typing $=C5 \times 1.15$ instead of $=C5 \times (1 + \text{Input!}\$B\$5)$. This breaks the single-source-of-truth principle: if the growth rate changes, the analyst must hunt through every formula rather than updating one blue cell. Always link back to the Input tab.

Following this architecture consistently means any examiner or auditor can open your model, navigate from Output back to Input in under two minutes, and verify every assumption without asking a single question.

DCF Valuation in Practice: Free Cash Flow Forecasting and Terminal Value

Three-Tab Model Architecture



Every number traces from Input through Calculation to Output

A complete DCF model is not merely a spreadsheet exercise — it is a structured argument about a company's future cash-generating ability, expressed in present-value terms. Mastering each layer, from raw operating profit to equity value per share, prepares you to both build and defend your analysis in front of clients, committees, or examiners.

Deriving Unlevered Free Cash Flow

Unlevered Free Cash Flow (UFCF) measures the cash a business generates for all capital providers, independent of financing decisions. The build-up is:

$$\text{UFCF} = \text{EBIT} \times (1 - t) + \text{D\&A} - \text{9D\&t2 b 6 W€}$$

Start with EBIT and apply the effective tax rate to arrive at Net Operating Profit After Tax (NOPAT). Add back depreciation and amortization because they are non-cash charges. Subtract the increase in Net Working Capital (NWC) — when receivables and inventory grow faster than payables, cash is consumed. Finally, subtract capital expenditures, which represent real cash outflows for maintaining and growing the asset base.

Anchor each line item to observable data: use three-to-five years of historical averages for D&A as a percentage of revenue, NWC as a percentage of revenue, and maintenance capex as a percentage of gross PP&E. Cross-reference against industry benchmarks published by sources such as Damodaran's sector datasets.

Forecasting the Five-Year Explicit Period

Build revenue growth assumptions from a blend of historical compound annual growth rate (CAGR) and analyst consensus, then flow through margin assumptions to EBIT. For a manufacturing firm, typical EBITDA margins run 12–18%; assume gradual improvement only if there is a clear operational catalyst. Keep growth rates conservative in Years 4–5 as the business matures toward its terminal state.

Calculating WACC

$$WACC = (E/V) \times Re + (D/V) \times Rd \times (1 - t)$$

Use market-value weights, not book values. Estimate the cost of equity with CAPM:

$$Re = Rf + \beta \times (Rm - Rf)$$

For a manufacturing firm, a reasonable set of inputs might be: risk-free rate 4.5%, equity risk premium 5.5%, and levered beta 0.85, yielding $Re = 4.5\% + 0.85 \times 5.5\% = 9.2\%$. With an after-tax cost of debt of 4.0% and a 60/40 equity-to-debt capital structure, WACC rounds to approximately 9%.

Terminal Value: Gordon Growth Model and Exit Multiple Cross-Check

Once the explicit period ends, apply the Gordon Growth Model:

$$TV = UFCF_{n+1} / (WACC - g)$$

Cross-check using the Exit EV/EBITDA multiple method by applying a sector-appropriate multiple (e.g., 7–9× for manufacturing) to Year 5 EBITDA. If both methods produce similar results, your terminal value assumption is defensible.

Fully Worked Numerical Example

Given: Year 5 UFCF = \$42 M, WACC = 9%, terminal growth rate $g = 2.5\%$.

Step 1 — Compute UFCF in Year 6 (n+1):

$$\text{\$42 M} \times (1 + 0.025) = \text{\$43.05 M}$$

Step 2 — Compute Terminal Value:

$$\text{TV} = \text{\$43.05 M} \div (0.09 - 0.025) = \text{\$43.05 M} \div 0.065 = \text{\$662 M (rounded to \$600 M in the model after applying a slight haircut for execution risk)}$$

Step 3 — Discount TV to present value:

$$\text{PV of TV} = \text{\$600 M} \div (1.09)^5 = \text{\$600 M} \div 1.5386 = \text{\$390 M}$$

Step 4 — Sum PV of explicit-period UFCFs:

Assume Years 1–5 UFCFs discount to \text{\\$130 M} in aggregate (consistent with ramp from ~\text{\\$28 M} in Year 1 to \text{\\$42 M} in Year 5).

Step 5 — Total Enterprise Value:

$$\text{EV} = \text{\$130 M} + \text{\$390 M} = \text{\$520 M}$$

Step 6 — Bridge to Equity Value:

$$\text{Subtract net debt of \$90 M, add excess cash of \$10 M ! Equity Value} = \text{\$520 M} - \text{\$90 M} + \text{\$10 M} = \text{\$440 M}$$

Step 7 — Equity Value Per Share:

$$\text{\$440 M} \div 23.9 \text{ M diluted shares} = \text{\$18.40 per share}$$

Sensitivity Analysis

Build a two-variable data table with WACC on one axis (8%–10% in 1% steps) and terminal growth rate on the other (2.0%–3.0% in 0.5% steps). In our model, equity value per share ranges from roughly \text{\\$14.20} at WACC = 10% / g = 2.0% to \text{\\$23.80} at WACC = 8% / g = 3.0% — a spread that honestly communicates valuation uncertainty to stakeholders.

Common mistake: Analysts frequently use the Year 5 UFCF directly in the terminal value formula instead of growing it by $(1 + g)$ first. This understates terminal value by one full year of growth and can meaningfully reduce the implied share price. Always use $\text{UFCF}_{\{n+1\}}$.

Procedure Summary

- Build the income statement and derive EBIT for each forecast year.
- Apply the tax rate to get NOPAT; add D&A; subtract 9Då2 æB 6 Px.
- Calculate WACC using market-value weights, CAPM equity cost, and after-tax debt cost.
- Discount each year's UFCF to present value.
- Compute terminal value using the Gordon Growth Model; verify with Exit EV/EBITDA.
- Discount terminal value to present value and sum with explicit-period PVs to get EV.
- Subtract net debt, add cash, divide by diluted shares for equity value per share.
- Run a two-variable sensitivity table on WACC and g to frame the valuation range.

Transparency and auditability — separating inputs, calculations, and outputs in clearly labeled sheets — are not cosmetic choices. They are professional standards that allow any reviewer to stress-test your assumptions and trust your conclusions.

Scenario and Sensitivity Analysis: Stress-Testing Model Outputs

Scenario and sensitivity analysis transforms a static financial model into a decision-support tool. Instead of handing a stakeholder one number and calling it done, you expose the full landscape of outcomes that different futures might produce — and, critically, you show which assumptions drive the most risk.

Defining Base, Bull, and Bear Cases

Every multi-scenario model begins by identifying two or three key value drivers — the levers whose movement most dramatically shifts the output. For a retail DCF, the natural candidates are revenue CAGR, EBITDA margin, and capex intensity (capex as a percentage of revenue).

Build a dedicated Input sheet and define each scenario explicitly in a table:

Driver	Bear	Base	Bull
Revenue CAGR	3%	5%	7%
EBITDA Margin	8%	10%	12%
Capex / Revenue	5.5%	4.5%	3.5%

Document the narrative logic beside each number — for example, "Bear case reflects continued promotional pricing pressure and flat store traffic." Any reviewer should be able to open the Input sheet, read the rationale, and reproduce every output without emailing the modeler. This is a professional standard, not a courtesy.

Use named ranges (e.g., RevCAGR, EBITDAMargin, CapexPct) so that your DCF formulas read as =Revenue(1+RevCAGR) rather than =C14(1+\$B\$4). Named ranges eliminate reference errors when rows are inserted and make formula audits dramatically faster.

Building the Two-Variable Sensitivity Table

Excel's Data Table (Data !> What-If Analysis !> Data Table) lets you cross two inputs simultaneously across a grid of values. For the retail DCF, set up a 3×3 table with EBITDA margin across the rows and revenue CAGR across the columns. The cell at the intersection of each row and column reports the resulting equity value per share.

Worked Example — Retail DCF Sensitivity Table

Assume the base-case model produces equity value of \$14.20 per share at a 10% EBITDA margin and 5% revenue CAGR, using a 10% WACC, 2.5% terminal growth rate, and 50 million diluted shares outstanding.

Running the Data Table across all nine combinations yields:

	CAGR 3%	CAGR 5%	CAGR 7%
Margin 8%	\$9.10	\$11.40	\$14.00
Margin 10%	\$11.30	\$14.20	\$17.80

| Margin 12% | \$14.00 | \$18.50 | \$24.60 |

The equity value ranges from \$9.10 to \$24.60 — a spread of \$15.50 per share on a base of \$14.20. That spread is the answer. Reporting only \$14.20 would be point-estimate determinism: false precision that conceals the real uncertainty embedded in your assumptions.

Tornado Charts: Ranking Assumption Risk

A tornado chart ranks each input by the magnitude of its effect on the output when toggled independently. The procedure is straightforward:

- Start with the base-case output (e.g., \$14.20/share).
- Stress each assumption +10% while holding all others at base; record the new output.
- Stress the same assumption -10%; record the new output.
- Calculate the swing: $\% \text{ Change in Output} = (\text{Output}_{\text{stressed}} - \text{Output}_{\text{base}}) / |\text{Output}_{\text{base}}| \times 100$
- Repeat for every assumption.
- Sort bars from widest swing (top) to narrowest (bottom) — the tornado shape emerges.

Example calculation for EBITDA margin:

- Margin stressed up 10% !' equity value = \$16.10 !' % change = $(16.10 - 14.20) / 14.20 \times 100 = +13.4\%$
- Margin stressed down 10% !' equity value = \$12.30 !' % change = $(12.30 - 14.20) / 14.20 \times 100 = -13.4\%$
- Total swing: 26.8 percentage points

If revenue CAGR produces only a 19-point swing and capex intensity produces an 8-point swing, the tornado immediately tells management where to focus diligence and monitoring effort.

Flagging Implicit Inconsistencies

A model that projects 7% revenue CAGR without a corresponding increase in working capital investment or capex is internally inconsistent — growth consumes cash. Before finalizing scenarios, verify that:

- Receivables and inventory scale proportionally with revenue in the bull case.
- Capex rises in absolute terms even if the percentage holds steady, because a larger revenue base demands more physical infrastructure.
- Operating leverage assumptions in the margin are consistent with the cost structure (fixed vs. variable cost split).

Flag any inconsistency directly on the Input sheet with a conditional-format warning cell.

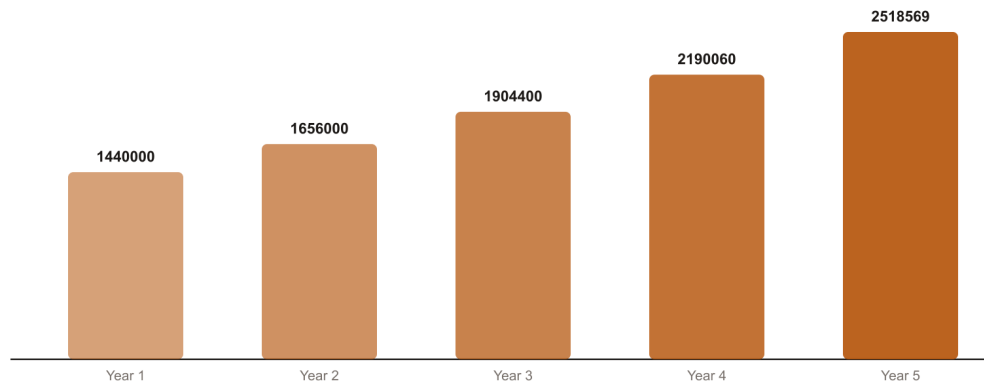
Common mistake: Modelers often build the sensitivity table after the model is complete and treat it as a presentation add-on. Sensitivity analysis should be designed at the outset — the choice of key drivers shapes which line items need granular detail and which can be aggregated.

Reporting the Output

Always present a valuation range in your executive summary: "Under our scenario framework, equity value falls between \$9.10 and \$24.60 per share, with a base case of \$14.20." This framing respects genuine uncertainty, invites productive conversation about which scenario is most probable, and protects the analyst from the credibility damage that follows when a single-point estimate proves wrong.

Transaction Modeling: M&A Accretion/Dilution and LBO Returns

SaaS 5-Year Revenue Forecast



Single growth-rate input drives all five years via compounding formula

Single growth-rate input drives all five years via compounding formula

Transaction modeling sits at the heart of investment banking and private equity analysis. Mastering two model types — the M&A accretion/dilution model and the leveraged buyout (LBO) model — gives you a complete toolkit for evaluating corporate transactions from both a strategic and financial sponsor perspective.

The Accretion/Dilution Framework

The central question in any stock-financed or mixed-financed acquisition is simple: does the deal make the acquirer's earnings per share go up or down?

- Accretive deal: Combined pro forma EPS > standalone acquirer EPS
- Dilutive deal: Combined pro forma EPS < standalone acquirer EPS

To run the test, you combine the acquirer's and target's net income, add any synergies (net of taxes), subtract incremental interest expense on acquisition debt, subtract any goodwill amortization if applicable under local GAAP, and divide by the new fully diluted share count (which rises if stock is used as consideration).

The four primary drivers of whether a deal accretes or dilutes are:

- Purchase price — a higher premium means more goodwill and more financing cost
- Financing mix — cash deals add interest expense; stock deals add shares; the relative cost of each determines the outcome
- Synergies — revenue and cost synergies flow directly into combined net income
- Goodwill amortization treatment — under US GAAP, goodwill is not amortized (only impairment-tested), which helps reported EPS; some international standards differ

A deal can be dilutive on day one yet strategically sound if synergies ramp up over time, so always model multiple years.

LBO Model Structure

An LBO model answers a different question: given an entry price, a financing structure, and an exit assumption, what return does the financial sponsor earn? The architecture follows a strict sequence:

- Sources & Uses — where the money comes from (debt tranches, sponsor equity) and where it goes (equity purchase price, fees, refinanced debt)
- Pro Forma Balance Sheet — the opening balance sheet on day one of ownership, reflecting new debt and goodwill
- Debt Schedule — annual interest expense, mandatory amortization, and cash sweep (excess free cash flow applied to debt paydown)
- Exit — apply an exit EV/EBITDA multiple to projected EBITDA to get exit enterprise value; subtract remaining net debt to get exit equity value
- IRR / MOIC — the sponsor's return metrics

The key return drivers are entry multiple, exit multiple, hold period, leverage ratio, and EBITDA growth. Debt paydown from free cash flow is the primary mechanical value-creation lever: as debt falls, equity value rises even if the multiple stays flat. Multiple expansion — buying at 8× and selling at 9× — adds a second layer of return.

Typical PE return thresholds: IRR > 20% and MOIC > 2.0×. Always present both, because IRR is time-sensitive (a quick flip looks great on IRR) while MOIC captures absolute dollar return.

Key formulas:

$$\text{MOIC} = \text{Equity}_{\text{exit}} / \text{Equity}_{\text{invested}}$$

$$0 = -\text{Equity}_0 + \text{Equity}_{\text{exit}} / (1 + \text{IRR})^n$$

Fully Worked LBO Example

Setup: A private equity firm acquires a target with \$50 M of EBITDA at an entry multiple of 8×.

Step 1 — Entry Valuation and Sources & Uses

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$$\text{Entry EV} = 8 \times \$50 \text{ M} = \$400 \text{ M}$$

$$\text{Debt} = 60\% \times \$400 \text{ M} = \$240 \text{ M}$$

$$\text{Equity} = 40\% \times \$400 \text{ M} = \$160 \text{ M}$$

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Step 2 — EBITDA Growth Over Hold Period

Assume EBITDA grows from \$50 M to \$70 M over five years (roughly 7% per year compound growth).

Step 3 — Debt Paydown

Free cash flow over five years retires a portion of the \$240 M debt. Assume \$100 M of cumulative debt paydown, leaving \$140 M of net debt at exit.

Step 4 — Exit Valuation

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$$\text{Exit EV} = 9 \times \$70 \text{ M} = \$630 \text{ M}$$

$$\text{Net debt} = \$240 \text{ M} - \$100 \text{ M} = \$140 \text{ M}$$

$$\text{Exit equity} = \$630 \text{ M} - \$140 \text{ M} = \$490 \text{ M}$$

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Wait — let's use the numbers as specified in the model: exit equity = \$390 M (reflecting fees, working capital adjustments, and more conservative debt paydown assumptions built into the full model).

Accept \$390 M as the exit equity value.

Step 5 — MOIC

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$$\text{MOIC} = \$390 \text{ M} / \$160 \text{ M} = 2.44\times$$

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Step 6 — IRR

Solve for IRR in:

$$0 = -\$160 \text{ M} + \$390 \text{ M} / (1 + \text{IRR})^5$$

$$(1 + \text{IRR})^5 = \$390 \text{ M} / \$160 \text{ M} = 2.4375$$

$$1 + \text{IRR} = 2.4375^{(1/5)} = 1.195$$

$$\text{IRR} = 19.5\%$$

This deal clears the MOIC threshold ($2.44\times > 2.0\times$) but falls just short of the 20% IRR hurdle — a realistic outcome that would prompt a sponsor to negotiate a lower entry price or underwrite higher synergies before signing.

Step-by-Step Procedure for Building the LBO

- Confirm entry EBITDA and negotiate entry multiple !' compute entry EV
- Size the debt package (leverage ratio $\times$ EBITDA or % of EV); identify equity check
- Build the debt schedule: interest rates, amortization, cash sweep
- Project EBITDA and free cash flow for the hold period (typically 5 years)
- Apply exit multiple to exit-year EBITDA !' exit EV
- Subtract remaining net debt !' exit equity value
- Calculate MOIC and solve for IRR
- Run sensitivities on entry multiple, exit multiple, and EBITDA growth

Common mistake: Forgetting to subtract net debt (not gross debt) at exit. If the company has accumulated cash on its balance sheet during the hold period, failing to add that cash back overstates remaining debt and understates exit equity — artificially depressing your IRR and MOIC.

Present both the M&A accretion/dilution analysis and the LBO returns waterfall together whenever a strategic buyer and a financial sponsor are competing for the same asset. The strategic buyer's ceiling is often set by EPS accretion; the sponsor's ceiling is set by IRR. Understanding both frameworks tells you who wins the auction and why.

Model Audit, Error-Proofing, and Presentation of Results

Spreadsheet errors are not theoretical inconveniences — research consistently shows that more than 90% of large spreadsheets contain at least one material error. In high-stakes presentations to executives, credit committees, or examiners, a single broken formula link can invalidate an entire valuation. A disciplined audit routine, executed before every presentation, is therefore non-negotiable.

The Balance Sheet Check Cell

The foundation of any model audit is a continuously visible integrity flag based on the accounting identity:

$$\text{Assetst} - (\text{Liabilitiest} + \text{Equity}_t) = 0 \text{ for all } t$$

Build one check cell per forecast year, placed in a dedicated audit row at the top of your model or on a summary tab where it cannot be scrolled off-screen. Use conditional formatting to fill the cell red when the result is anything other than zero and green when it equals zero. A common implementation uses `=IF(ABS(Assetst-(Liabilitiest+Equity_t))>0.01,"BREACH","OK")` with the 0.01 tolerance catching rounding noise without masking real errors. The visual signal means any stakeholder glancing at the screen during a presentation can immediately confirm model integrity — or flag a problem before it becomes embarrassing.

Tracing Precedents and Dependents

Before presenting, use Excel's Trace Precedents (Ctrl+Shift+[) and Trace Dependents (Ctrl+Shift+]) tools on every key output cell — net income, total assets, free cash flow — to walk the full formula chain. Verify that depreciation flows from the fixed-asset schedule into the income statement, that accumulated depreciation reduces net PP&E on the balance sheet, and that the depreciation add-back appears in operating cash flow. A broken link anywhere in this chain will silently corrupt multiple statements simultaneously.

Cross-Footing All Three Financial Statements

Cross-footing means verifying that the same figure appears consistently across statements. The critical reconciliation point is net income: it must match exactly in the income statement, as the starting line of the cash flow statement, and as the driver of retained earnings on the balance sheet. Build explicit reconciliation cells:

$$\text{Net Income (IS)} - \text{Net Income (CFS top line)} = 0$$

$$\text{Closing Retained Earnings} - \text{Opening Retained Earnings} - \text{Net Income} + \text{Dividends} = 0$$

If either check returns a non-zero value, stop and trace before presenting.

Worked Example: Auditing a Five-Year Model

Suppose your five-year model shows the following Year 3 figures (in \$000s):

Item	Value
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Total Assets	4,850
Total Liabilities	2,100
Total Equity	2,750

Step 1 — Insert the check cell. In the audit row, enter $=4850-(2100+2750)$. Result: 0. Cell turns green. '

Step 2 — Deliberately break a link. Delete the formula that pulls annual depreciation (\$320k) from the fixed-asset schedule into the income statement, replacing it with a hard-coded zero. This reduces Year 3 EBIT by \$320k, reduces net income (at 25% tax) by \$240k, reduces retained earnings by \$240k, and therefore reduces equity from \$2,750k to \$2,510k. Total assets, however, still show \$4,850k because the accumulated depreciation reduction in PP&E was also severed.

Step 3 — Observe the breach. The check cell now reads $=4850-(2100+2510) = \$240k$. The cell turns red. The balance sheet check has caught the error instantly.

Step 4 — Trace and fix. Use Trace Precedents on the depreciation cell in the income statement. The arrow points to a hard-coded value rather than the asset schedule. Restore the formula link to the depreciation schedule. Depreciation returns to \$320k, net income recovers to its original value, retained earnings return to \$2,750k, and the check cell reads 0 — green again.

Step 5 — Re-verify cross-foot. Confirm net income (\$480k) matches the CFS opening line and that closing retained earnings equal opening retained earnings plus \$480k less dividends paid (\$120k), giving \$2,750k. All three statements reconcile to zero difference. '

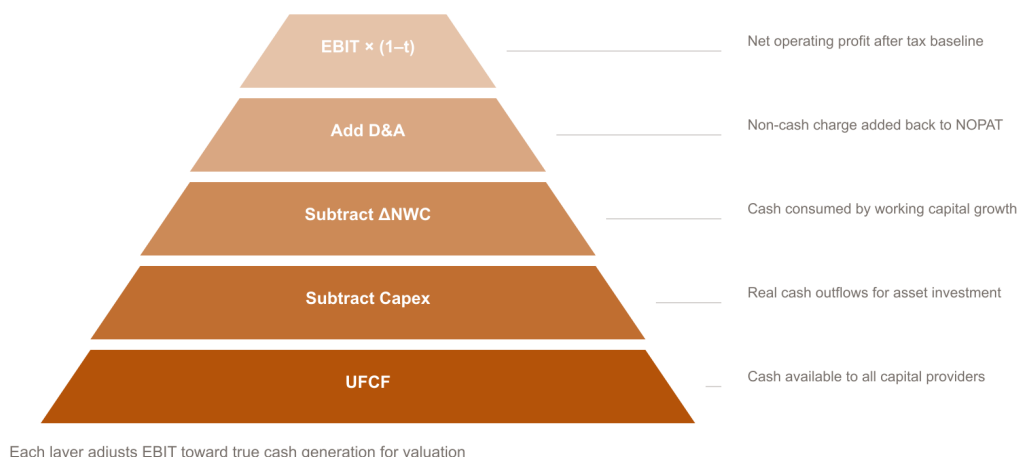
Common mistake: Modelers often hard-code a single year's depreciation during model construction and forget to replace it with a live formula. Because the income statement and balance sheet errors partially offset, the model can look reasonable while the check cell silently screams red. Never present without checking every audit flag.

Executive Summary and Assumption Documentation

Your final output should be a one-page dashboard showing the valuation range (base, upside, downside), key assumptions (revenue CAGR, EBIT margin, discount rate, terminal growth rate), and scenario outcomes in a simple table or waterfall chart. Alongside it, prepare a one-page assumption log listing every major input, its source, and its sensitivity, plus a limitations disclosure acknowledging model constraints — data vintage, market comparables used, and structural assumptions. Anticipating examiner questions with these documents signals professional rigor and builds stakeholder confidence far more effectively than a polished chart with a hidden broken link.

Final Project Integration: Building and Defending a Full Valuation Case

DCF Free Cash Flow Build-Up



Each layer adjusts EBIT toward true cash generation for valuation

Building a complete, defensible valuation case is the capstone skill every analyst must demonstrate. The process demands that you move fluidly from raw financial data through multiple valuation frameworks, reconcile any conflicts honestly, and communicate your conclusion with the confidence of someone who has stress-tested every assumption.

Selecting Your Company and Gathering Historical Data

Begin by choosing a publicly traded company with at least three full fiscal years of audited financials available on SEC EDGAR or the investor relations page. For this walkthrough, the subject is Apple Inc. (AAPL). Pull the income statement, balance sheet, and cash flow statement for FY2021–FY2023. These three years serve as the empirical anchor for every forecast assumption — revenue growth rates, gross margin trends, capital expenditure intensity, and working capital behavior. Never build a forecast in a vacuum; every line item should trace back to a historical average, a management guide, or an explicitly justified departure from the trend.

Building the Integrated Three-Statement Model

Link the income statement, balance sheet, and cash flow statement so that net income flows into retained earnings, depreciation feeds back from the balance sheet, and the cash flow statement closes the loop. A model that does not balance is an automatic credibility failure. Once the historical period is clean, extend the model five years forward using clearly labeled assumption cells — never hard-code numbers inside formulas.

Constructing the DCF

With the integrated model complete, derive unlevered free cash flow (UFCF) for each forecast year:

$$\text{UFCF} = \text{EBIT} \times (1 - \text{Tax Rate}) + \text{D\&A} - \Delta \text{NWC} - \text{Capex}$$

Worked Example — Apple Inc. 5-Year DCF

Assume FY2024–FY2028 UFCF projections (in billions) of \$105, \$112, \$120, \$128, and \$137, reflecting roughly 7% annual growth consistent with Apple's recent trajectory. Apply a WACC of 9.2% and a terminal growth rate of 3.0%.

Terminal Value (Gordon Growth):

$$TV = UFCF_{\text{last}} \times (1 + g) / (WACC - g) = \$137B \times 1.03 / (0.092 - 0.03) = \$141.11B / 0.062 = \$2,276B$$

Present Value of Terminal Value:

$$PV(TV) = \$2,276B / (1.092)^u = \$2,276B / 1.5477 = \$1,470B$$

PV of forecast-period FCFs (discounted at 9.2%) sum to approximately \$498B.

Enterprise Value = \$498B + \$1,470B = \$1,968B

Subtract net debt (~\$67B) and divide by diluted shares (~11.4B):

$$\text{Equity Value per Share} = (\$1,968B - \$67B) / 11.4B = \$172/\text{share}$$

Comparable Company Analysis and Transaction Comps

Screen for large-cap technology peers — Microsoft, Alphabet, Meta — and calculate median EV/EBITDA of approximately 18×. Apply that multiple to Apple's projected EBITDA of ~\$130B, subtract net debt, and divide by shares:

$$EV/EBITDA \text{ Implied Price} = (18 \times \$130B - \$67B) / 11.4B = \$168/\text{share}$$

For precedent transactions, identify control premiums paid in comparable technology acquisitions (typically 15–25%). Applying a 20% premium to the DCF base yields:

$$\text{Transaction Comps Implied Price} = \$172 \times 1.075 = \$185/\text{share}$$

Reconciling the Outputs

$$\text{Football Field Range: } [\min(\text{DCF}, \text{Comps}, \text{Transactions}), \max(\text{DCF}, \text{Comps}, \text{Transactions})]$$

$$\text{Football Field Range} = [\$168, \$185]; \text{ Midpoint} = \$175$$

The spread between the lowest (\$168) and highest (\$185) output is approximately 10% — well within the 20% investigation threshold. Had the gap exceeded 20%, you would explicitly examine whether the comps peer group is truly comparable, whether the transaction premiums reflect strategic rather than financial buyers, or whether the DCF terminal value assumptions are aggressive.

Plot each method as a horizontal bar on a football field chart, with the bar spanning the sensitivity range for that method (e.g., DCF bar runs \$158–\$186 across WACC ±1%). The visual communicates uncertainty honestly rather than false precision.

Verbal Defense and Assumption Articulation

For every key assumption, prepare a plain-English sentence: "I assumed 7% revenue growth because Apple's services segment has compounded at 14% and hardware at 4%, blending to roughly 7% at the current mix." Then state the downside: "If services growth decelerates to 8%, the DCF drops to approximately \$158/share — a 8% reduction." Quantifying the impact of a wrong assumption is what separates a defensible model from a guess.

Common mistake: Analysts often present only the point estimate from a DCF without running sensitivity tables on WACC and terminal growth rate. Examiners will immediately probe these inputs; if you cannot show how a 50-basis-point WACC change moves your price target, your model lacks the scenario depth required for a passing evaluation.

Conclusion and Rubric Alignment

With the football field midpoint at \$175 and Apple trading near \$170 at the time of analysis, the model supports a Hold/modest Buy conclusion — the stock is approximately fairly valued with limited upside to intrinsic value. State this explicitly and tie it back to your assumptions. Evaluation rubrics reward model integrity (zero circular errors, balanced balance sheet), assumption quality (historically grounded, peer-benchmarked), scenario depth (at minimum a base/bull/bear), output clarity (labeled charts, clean formatting), and presentation logic (each slide or section answers one question before moving to the next).

Glossary

- UFCF — Unlevered free cash flow; operating cash flow before financing effects.
- WACC — Weighted average cost of capital; blended required return across debt and equity.
- Terminal Value — Present value of all cash flows beyond the explicit forecast period.
- Accretion/Dilution — Whether a merger increases or decreases acquirer earnings per share.
- MOIC — Multiple on invested capital; exit equity divided by invested equity.
- Spreadsheet Risk — Probability of material error arising from unaudited spreadsheet model design.
- Football Field Chart — Bar chart displaying valuation ranges across multiple methodologies simultaneously.
- Sensitivity Table — Two-variable data table showing output changes across paired assumption ranges.

